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Day 4 — Task 1: Improve Pose Detection Accuracy

Pose Detection

Pose detection is a computer vision technique that identifies and tracks the position of human body parts in an image or video. It works by detecting key points on the body called landmarks such as shoulders, elbows, wrists, hips, knees, and ankles. These landmarks are then connected to form a skeleton-like structure that represents the human pose.

Why Accuracy Needs Improvement

Out of the box, MediaPipe works well but there are situations where it struggles. Poor lighting causes the model to miss landmarks or assign low visibility scores. If the person is too far from the camera or partially out of frame, fewer landmarks will be detected. A cluttered background can confuse the detector. The default settings use a balanced mode that prioritises speed over accuracy, so switching to accuracy mode gives better results.

How to Improve Accuracy

-The most important change is setting model complexity to `2` instead of the default `0`. This uses the most powerful version of the model and detects landmarks more precisely especially on difficult images.
-Raising the detection confidence threshold to `0.7` means the model only reports a detection when it is at least 70 percent confident, reducing false detections.
-Good lighting matters a lot. Even, bright front-facing light helps the model see body outlines clearly. Avoid backlight or harsh shadows.

-Camera position should be straight-on and at full body height if possible. Side angles or extreme top-down angles reduce accuracy significantly.

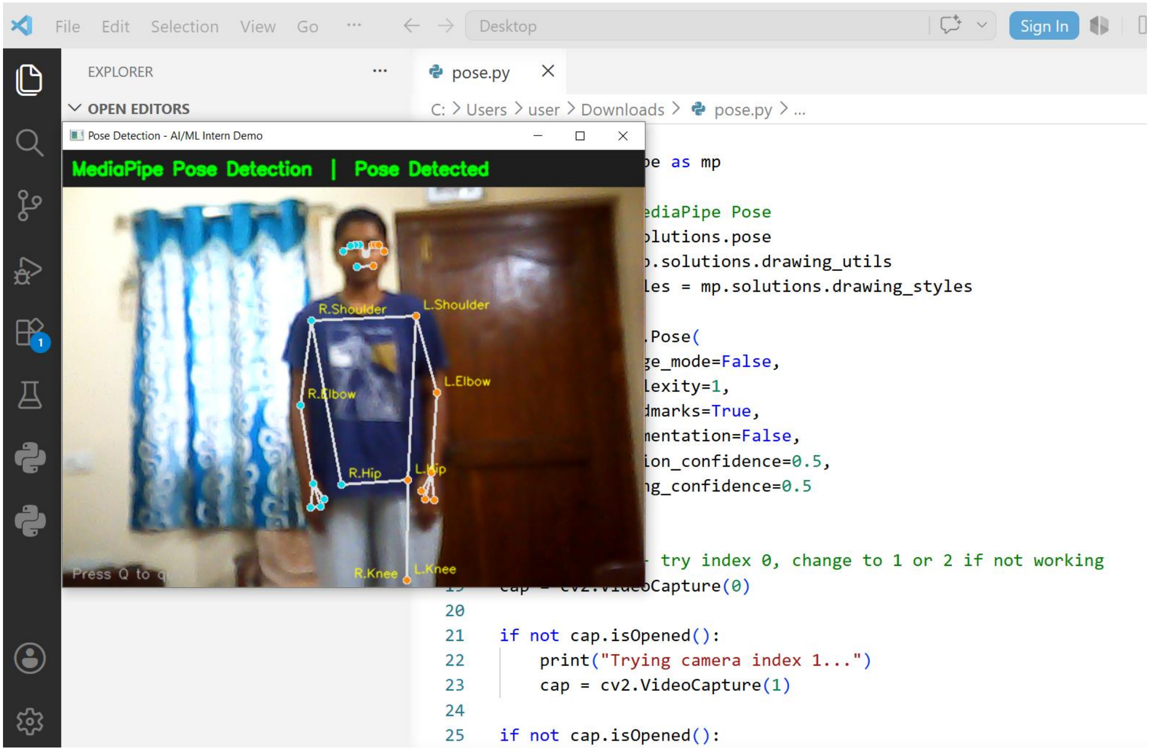
-A plain or contrasting background helps the model distinguish the person from the surroundings more easily.

Key Landmarks for Virtual Try-On

Not all 33 landmarks are needed for clothing overlay. The most important ones are the left and right shoulders, left and right elbows, left and right wrists, and left and right hips. The distance between the two shoulder landmarks in pixels is called the shoulder width and is used to scale the garment to the correct size for that person.

Ouptput Observation

Today's pose detection output is significantly more accurate compared to yesterday. The improved accuracy is mainly due to better lighting conditions during testing. With proper front-facing light, the model was able to detect all 33 landmarks clearly with high visibility scores above 0.9 for key joints like shoulders and hips.



Task 2: Research on AI Virtual Fitting System

I researched how modern AI based virtual try-on systems work.
A virtual fitting system allows a person to see how a piece of clothing would look on them without physically wearing it. This is done entirely using computer vision and deep learning models. I studied the complete workflow of how these systems are built and what each stage does.

What is a Virtual Try-On System

A virtual try-on system takes two inputs , a photo of a person and a photo of a clothing item and produces an output image where the person appears to be wearing that clothing. It does not simply paste the clothing on top of the person. Instead it understands the body shape, adjusts the clothing to fit naturally, and blends it realistically.

Virtual Try-On Workflow

Stage 1 — Person Parsing

The body is divided into regions like face, torso, arms and legs.
Each pixel is labeled so the system knows what part of the body it is.

Stage 2 — Cloth Segmentation

The clothing item is separated from its background so only the garment remains as a clean image ready for the next stage.

Stage 3 — Warping

The flat garment image is stretched and reshaped to match the person's body pose and shape so it fits naturally.

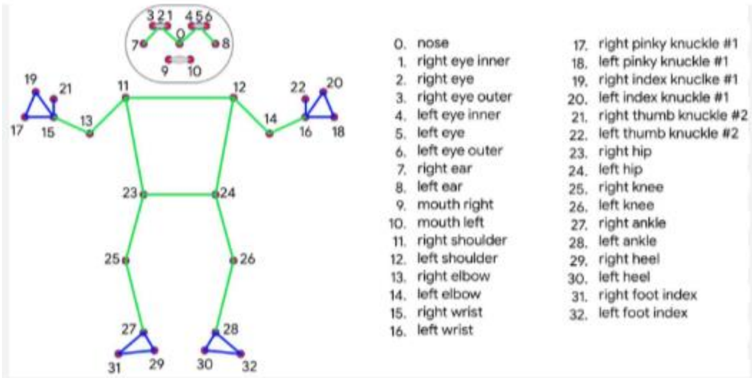
Stage 4 — Blending

The warped garment is merged onto the person's image using alpha blending to make the final output look realistic.

Pose Estimation

Pose estimation is the foundation of the entire virtual try-on system. Before any garment can be placed on a person the system needs to know exactly where the body is and how it is positioned. Pose estimation detects key body joints called landmarks such as shoulders, elbows, wrists, and hips.

I used MediaPipe Pose for this task which detects 33 landmarks from a single image.



Segmentation

Segmentation divides the image into meaningful parts by classifying every pixel. For virtual try-on, the person is separated from the background using a body mask. This ensures the garment only appears on the body and not on the background. MediaPipe Selfie Segmentation is used for this which returns a confidence mask where 1 means person and 0 means background.

Garment Overlay

Garment overlay is placing the clothing item on the person's image after pose detection is done. The shoulder coordinates detected by

MediaPipe are used to find the position and the shoulder width is used to scale the garment to the correct size. The garment is then placed at that position using alpha blending so it looks natural on the person.

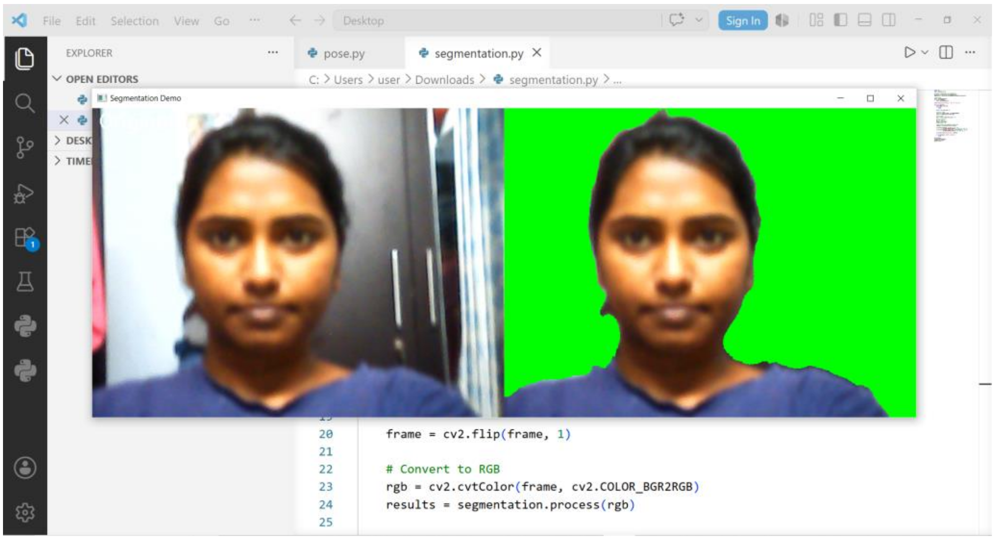
Task -3

MediaPipe Selfie Segmentation — Background Removal

MediaPipe Selfie Segmentation is a pre-trained AI model that separates the human body from the background in real time. It processes each video frame and generates a segmentation mask where each pixel is classified as either a person or background. A threshold of 0.5 is applied to the mask — pixels above 0.5 are considered as person and pixels below are considered as background.

How it Works

- Camera captures a video frame
- Frame is converted from BGR to RGB format
- MediaPipe processes the frame and generates a segmentation mask
- Mask values above 0.5 are kept as the person
- Remaining pixels are replaced with a chosen background color
- Output is displayed in real time



What I Observed

- The model successfully detected and separated the human body from the background in real time
- Left side of the screen showed the original webcam feed
- Right side showed the person with green background replacing the original background
- The segmentation worked smoothly even with movement

Task-4 : How AI Recommends Clothing

Overview

AI clothing recommendation is a system that suggests outfits to a user based on their personal preferences, body type and external conditions like weather and occasion. Instead of manually browsing through thousands of items the AI filters and ranks the most suitable options automatically.

How It Works

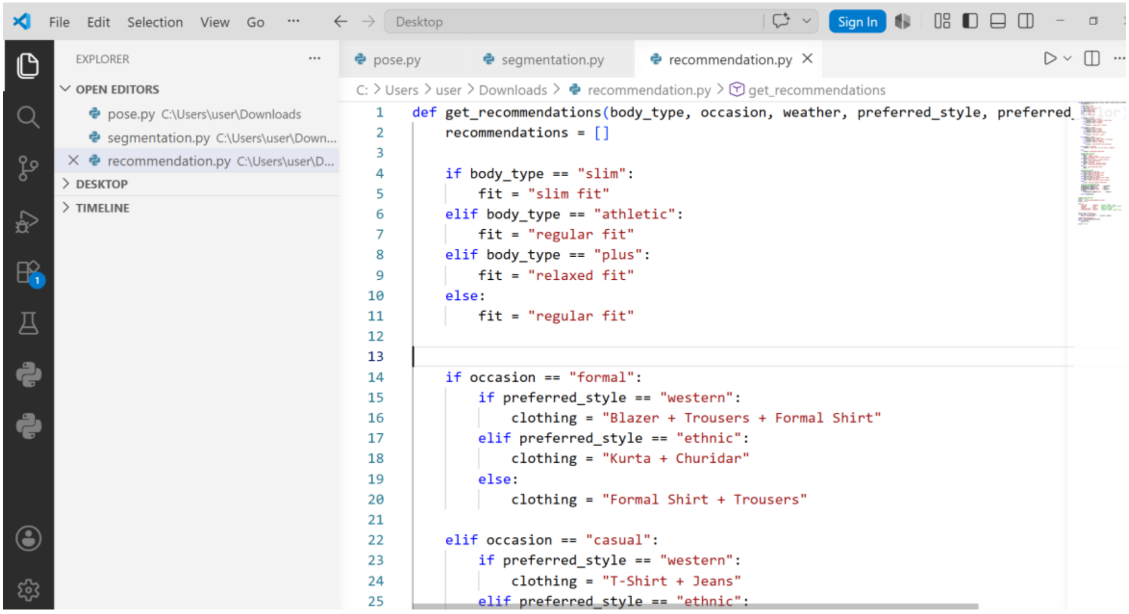
The system takes four inputs from the user — body type, occasion, weather and style preference. Based on these inputs a logic engine applies a set of conditions to determine the most suitable outfit, fabric, fit and color combination.

- Body type determines the fit — slim fit for slim body, relaxed fit for plus size
- Occasion determines the clothing category — formal, casual, party or sports
- Weather determines the fabric — cotton for hot weather, wool for cold weather
- Style preference determines the outfit style — western or ethnic

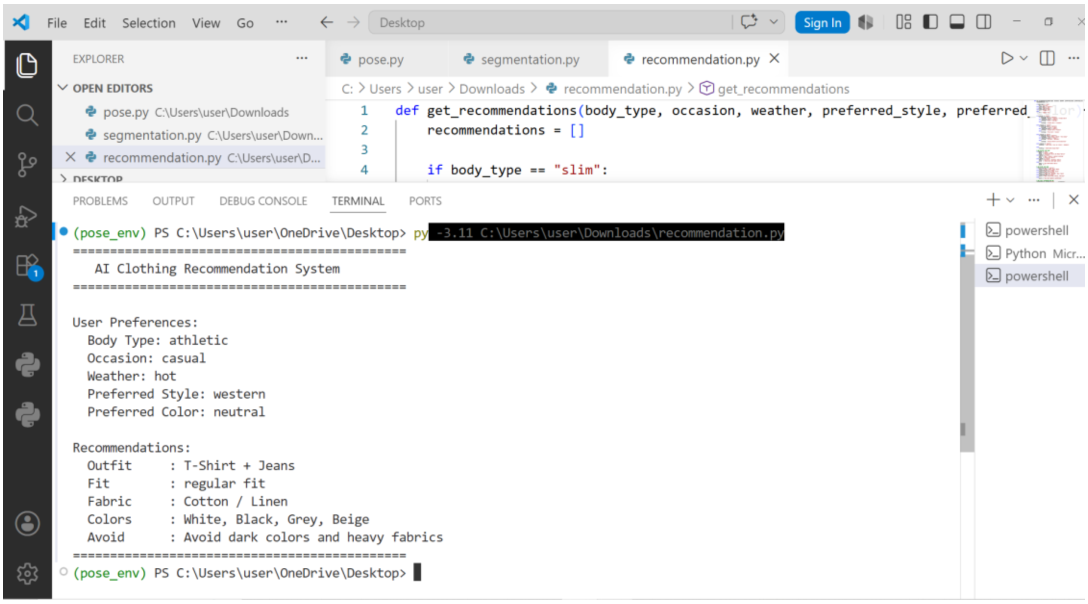
Why AI is Better Than Manual Search

A normal user searching online has to apply multiple filters manually and still may not find the right fit. AI combines all these conditions simultaneously and gives a single accurate recommendation in seconds. As the user interacts more with the system the AI learns their preferences and becomes more accurate over time.

Code:



Output:



Task-5

MediaPipe:

- Developed by Google as a cross platform machine learning framework
- Uses BlazePose model which detects 33 body landmarks including face, hands and full body
- Very fast and runs in real time on normal CPU without needing a GPU
- Easy to install with just one pip command and integrates simply with Python and OpenCV
- Works best for single person detection in controlled environments
- Ideal for mobile apps, real time applications and beginner level projects
- Best choice for virtual try-on systems where one person is detected at a time

OpenPose:

- Developed by Carnegie Mellon University as an open source pose detection system
- Uses Part Affinity Fields technique which detects 25 body landmarks
- Can detect multiple people simultaneously in a single frame
- Requires a GPU for real time performance making it heavy on hardware
- Installation is complex and setup takes more time compared to MediaPipe
- More accurate in crowded scenes and multi-person environments
- Best suited for academic research and large scale projects